

Security Overview

OLLI Digital Explorers SIG

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Agenda

- Show & Tell
- Security
 - What is cyber security?
 - What is my risk exposure?
 - Cyber security “toolbox”
- Choose the topic for our next meeting



- You **can reduce** security risks; you **can't completely eliminate** the possibility that something bad can happen.
- There are no “tried and true” methods to quantify the level/degree of risk you are exposed to. E.g., “If you do x , there is an $n\%$ chance you will get hacked or get malware on your device.”
- I am intentionally being as **transparent** as possible about cyber security threats and risks.
- Don't go off half cocked. It's easy to make mistakes that result in problems you'll want/need to immediately fix. (Trust me. I've locked myself out of accounts more times than I want to admit.)

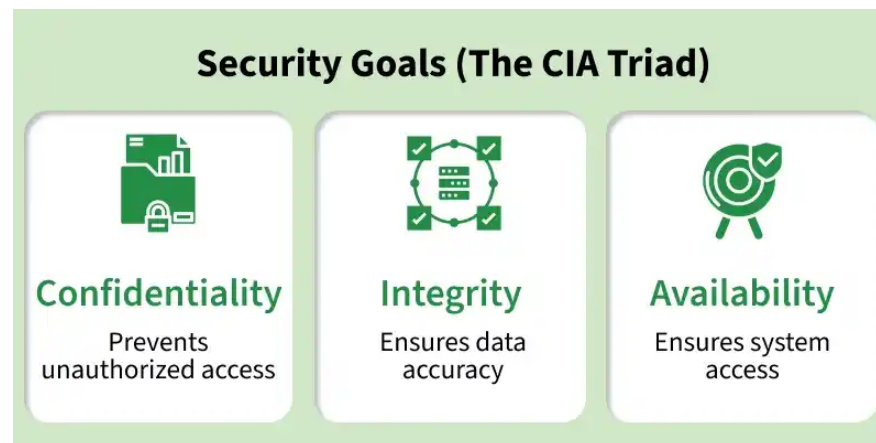
Human error is consistently recognized as the weakest link in cybersecurity. — cyberpeace.org

- Make a plan. Be prepared to invest time in executing your plan.
- Sustaining your desired level of security into the future involves routinely reviewing & maintaining your plan.

What is Cybersecurity

“Cybersecurity is the practice of protecting digital systems, networks, cloud environments, and data from unauthorized access, misuse, or disruption. It’s a constantly evolving discipline that integrates technology, policy, and human expertise to defend the confidentiality, integrity, and availability of critical assets across the modern digital landscape.”

— *geeksforgeeks.com*



What are we protecting?

- Devices
- Email accounts
- Site accounts
- Personally identifiable information (PII)
 - legal name, date of birth, home address, phone number, social security number, passport number, drivers license number
- Files stored in the cloud / shared with others digitally

Threats

Bad actors often use different attacks simultaneously to perpetrate their crimes. This is called a **multi-vector** attack.

Malware software can change over time—making it extremely difficult for anti-virus software vendors to detect malware. Malware that changes in the way is referred to as **polymorphic**.

- Malware: viruses, worms, ransomware, spyware, botnets, adware, rootkits, keyloggers
- Social engineering (i.e., phishing or vishing)
- Scams
- Identify theft and abuse
- Credential theft and abuse
aka “account hijacking”
- Emerging: agentic AI
- Backdoor attacks
- Man-in-the-middle attacks
- HTML smuggling attacks
- Eavesdropping attacks
aka “network sniffing” and “snooping attacks”
- Privilege escalation attacks
- Spoofing attacks
- Cookie theft
- Tampering attacks
- Denial-of-service (DDoS) attacks

Who perpetrates cybersecurity attacks?



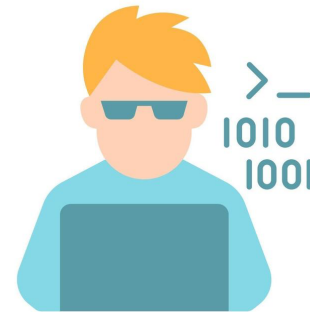
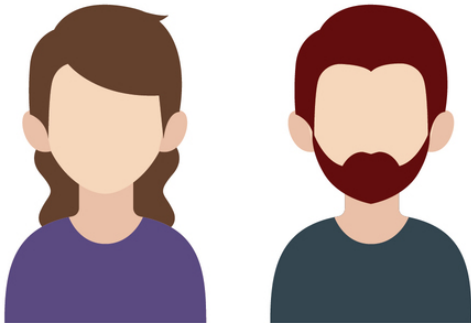
- **Hacker:** individuals who use their technical skills to gain unauthorized access to systems.

There are various kinds of hackers. Some common terms for different kinds of hackers are black hat hackers, gray hat hackers, crypto jackers, script kiddies, hacktivists.

To be clear: **hackers break the law and commit crimes.**

- **Hacker groups (aka hacker gangs)**
- **Ransomware groups**
- **Cyber mercenaries**
- **Nation-state sponsored actors**

Who do I **trust**?



IT Engineer



Software Developer



Cyber Security Researcher



What's my level of risk?

Your level of risk is a combination of (1) the kinds of things you do and (2) how frequently you do them. Frequency has more weight than the kinds of thing you do in this equation. I.e., the more you do the following thing, the higher the your odds of “getting hacked” are.

- Number of weak passwords and/or reused passwords
- Number of accounts you create
- Number of different sites you explore, pages you visit, links you click, etc.
- Number of email/text attachments you open and the number of files you download and open
- Number of apps you install
- Amount of time you spend using public Wi-Fi

Cyber security experts considered people's persistence using weak/reused passwords and falling for phishing texts/email/phone calls to be the two greatest threats to cyber security across the digital domain writ large.

Your “Toolbox”

- Strong and unique passwords, 2FA/MFA and passkeys
- Password managers
 - To use passkeys with an account, you have to use a password manager
- File encryption
- Automatic software updates

Note: I have a few apps that don’t “participate” in Apple’s automatic updates on macOS, so I had to develop a habit of routinely checking those apps to see if updates are available.

- Anti-virus software
 - macOS and Windows OS both include anti-virus software; they provide sufficient protection for most people.
 - iPhones and iOS do not include anti-virus software. There are antivirus apps available in the App Store (most if not all are subscription based). There have been extremely few cases of iPhones and iPads viruses; there are a few notorious cases of spyware on iPhones and iPads. For every case I know of, the associated bad actors targeted very specific kinds people (e.g., politicians, diplomats, journalists, dissident activists, etc.)
- Settings
 - There are system **and** app-level settings related to security. In my experience with Apple products, security related system level settings are usually set for maximum protection by default. The same is true for Apple apps. Apple’s ecosystem is such that apps created by others companies should have similar default values for any security settings they support—but to be honest, I don’t trust that will always be the case so I usually check when I install any non-Apple apps.
- Virtual Private Network (VPN) services

Your “Toolbox”

Tool	What it protects
Two-factor/Multi-factor Authentication	• Accounts
Passkeys	• Accounts
Password Managers	• Accounts
File Encryption	• Files stored in the cloud
Settings <i>System- and app-level</i>	• Inadvertently visiting suspicious/ unsecure sites • Device from malware
Automatic Software Updates	• Apps and Operating System
Anti-Virus Software	• Device from malware
Virtual Private Network (VPN)	• Use of public Wi-Fi

My Digital Garden

I have to spend time maintaining my devices and what I have on them (e.g., files/data and apps). I think of my digital assets as a garden and follow a plan for maintaining my garden. The plan is based on a strategy that keeps me confident I have the level of security & privacy that matters to me.



- My “**best practices**” strategy focuses on
 - Maximizing the value/lifetime I get from my devices
 - Enabling me to use my devices as easily & efficiently as possible
 - Securing my digital assets
 - Maintaining a level of privacy that I’m comfortable with
- I organize my plan covers three key areas (to keep me “sane”):
 - General housekeeping
 - Security
 - Privacy

Digital Garden: Security

I use public Wi-Fi about once a week. I use a VPN service when I use public Wi-Fi.

Note: VPNs can increase your degree of privacy in addition to reducing your security risks.

- I keep an inventory of assets (accounts, apps, & confidential files)
- Very detailed inventory (i.e, a spreadsheet) of email and site accounts
 - E.g., which site accounts have my phone number or credit card number, what kind of 2FA/MFA I've set accounts to use, etc
- I have fairly specific “rules” related creating/configuring my accounts and how I organize my files
- **As long as I adhere to the “rules”, it takes very little time/effort to maintain the security aspect of my digital garden**

Devices	MacBook, iPhone, 2 iPads <i>MacBook is my <u>primary</u> device</i>
Email Accounts	• Apple: 3 • Google: 6 (I'm locked out of one) • Proton: 5
Site Accounts	• Total: 125 • Maximum security: 25 • Contain PII: 61 • Anonymous: 39
Files	I <u>only</u> put files/data that I want to sync between my devices in cloud storage. • Apple Apps • Contacts, Calendar, Photos, Notes, Passwords, Safari (bookmarks) • Bitwarden (a password manager)

Low hanging fruit

While there is a lot more to be said about the things you do to reduce your security risk, below are a some things you can consider doing now. I.e., the steps below involve knowledge/skills we covered in DESIG meetings earlier this year. The first two suggestions reduce your risk of getting hacked; the remaining suggestions reduce your risk of getting scammed and getting malware on your device(s).

- Review your passwords.
 - Identify and change any that are not unique or strong.
 - Turn on two-factor/multi-factor authentication (2FA/MFA) via text/email for accounts that support it.
A site that supports 2FA/MFA should give you a set of “backup codes” when you turn 2FA/MFA on. You need to **save a copy** of those codes. If you somehow forget/lose your password, you can use one of these backup codes to get into your account and reset the password.
- Review files you stored on your cloud drive.
 - Move files that contain sensitive/confidential information from your cloud drive to your local drive.
- **Never** open a file attached to an email or text message from an email address or phone number you do not recognize. The same goes for emails/texts that appear to be from a company/organization you do business/have an online account with. (Hackers impersonate companies/organizations!)
- **Never** take any actions suggested in a text/email from any business/organization. I.e., don’t “click this link” or “call this number”.
Instead if you have an online account with that business/organization, sign into that account and look for any notifications/indications there of any actions the business/organization is requesting/suggesting you take. Or call the business/organization and ask a customer service representative if text/email you received is from them.
- **Never** click on ads you see in your browser or apps you use. They can be scams. Viewing them can also lead to malware being installed on your device.
Instead if you see an advertisement you are interested in, use your preferred Internet search engine to find the associated business’s site and search the site for the advertised product you are interested in.

Next meeting topic

- If we want to continue with security, there are two additional top level topics:
 - How things work
E.g., what happens when you sign-in; the “flavors” of 2FA/MFA; encryption
 - How to use items in the toolbox
As well things to consider when picking exact tools (e.g., a password manager or a VPN provider)
- The “Software” section of the Tech Skills 101 course
 - There some of the items in the security toolbox are complex enough that it may beneficial to spend some time reviewing that material before we have a deeper discussion on how to use those tools.
- Other suggestions/ideas?